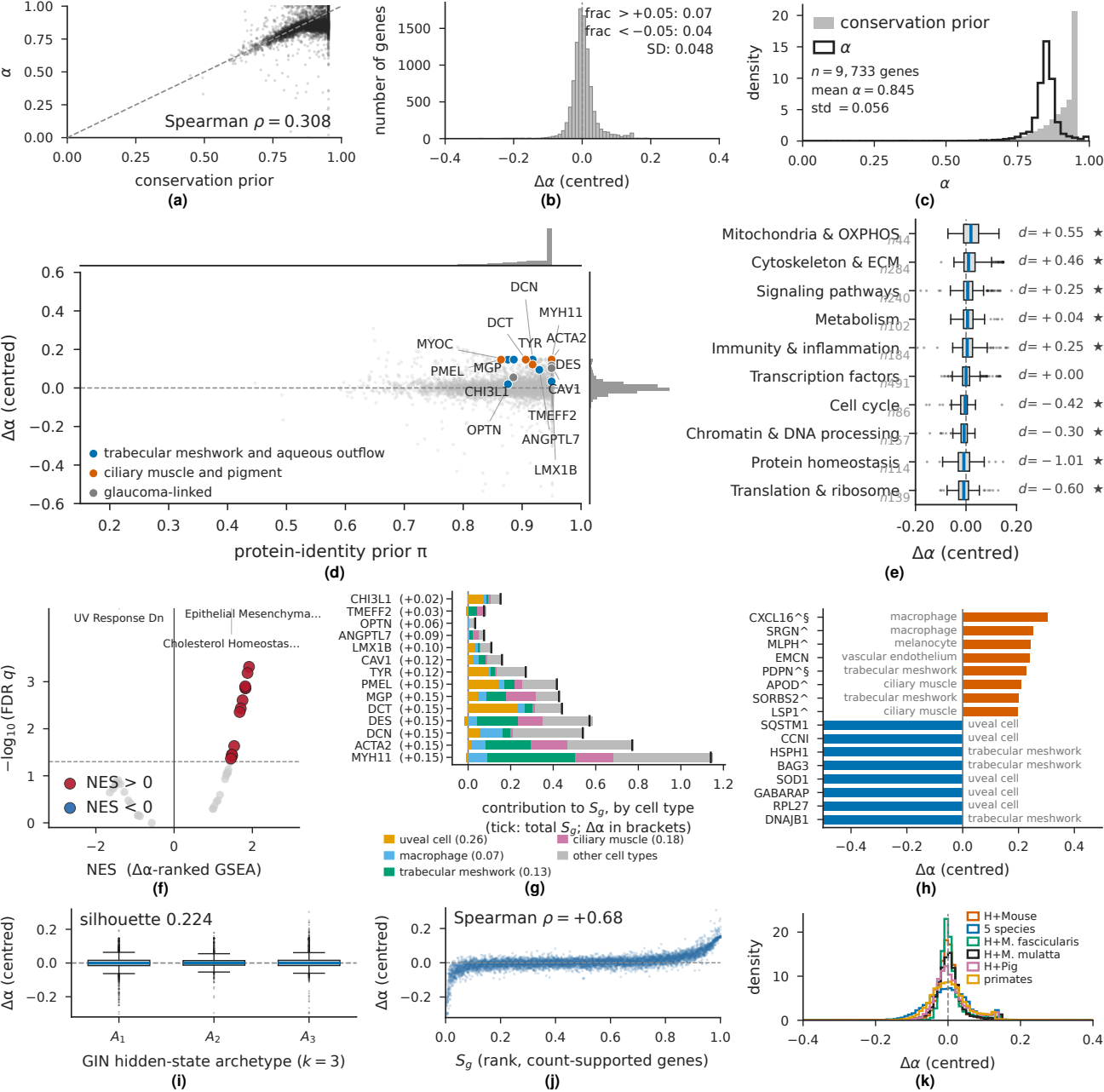

Spearman ρ = 0.308
conservation prior
α
(a)
frac > +0.05: 0.07
frac < −0.05: 0.04
SD: 0.048
number of genes
Δα (centred)
(b)
conservation prior
α
n = 9,733 genes
mean α = 0.845
std = 0.056
density
(c)
DCN
MYH11
DCT
ACTA2
TYR
MYOC
DES
PMEL
MGP
CAV1
CHI3L1
TMEFF2
OPTN
ANGPTL7
LMX1B
trabecular meshwork and aqueous outflow
ciliary muscle and pigment
glaucoma-linked
Δα (centred)
protein-identity prior π
(d)
Mitochondria & OXPHOS n44 d = +0.55 ★
Cytoskeleton & ECM n284 d = +0.46 ★
Signaling pathways n240 d = +0.25 ★
Metabolism n102 d = +0.04 ★
Immunity & inflammation n184 d = +0.25 ★
Transcription factors n491 d = +0.00
Cell cycle n86 d = −0.42 ★
Chromatin & DNA processing n157 d = −0.30 ★
Protein homeostasis n114 d = −1.01 ★
Translation & ribosome n139 d = −0.60 ★
Δα (centred)
(e)
UV Response Dn
Epithelial Mesenchyma...
Cholesterol Homeostas...
NES > 0
NES < 0
−log10(FDR q)
NES (Δα-ranked GSEA)
(f)
CHI3L1 (+0.02)
TMEFF2 (+0.03)
OPTN (+0.06)
ANGPTL7 (+0.09)
LMX1B (+0.10)
CAV1 (+0.12)
TYR (+0.12)
PMEL (+0.15)
MGP (+0.15)
DCT (+0.15)
DES (+0.15)
DCN (+0.15)
ACTA2 (+0.15)
MYH11 (+0.15)
contribution to S_g, by cell type
(tick: total S_g; Δα in brackets)
uveal cell (0.26)
ciliary muscle (0.18)
macrophage (0.07)
other cell types
trabecular meshwork (0.13)
(g)
CXCL16^§ macrophage
SRGN^ macrophage
MLPH^ melanocyte
EMCN vascular endothelium
PDPN^§ trabecular meshwork
APOD^ ciliary muscle
SORBS2^ trabecular meshwork
LSP1^ ciliary muscle
SQSTM1 uveal cell
CCNI uveal cell
HSPH1 trabecular meshwork
BAG3 trabecular meshwork
SOD1 uveal cell
GABARAP uveal cell
RPL27 uveal cell
DNAJB1 trabecular meshwork
Δα (centred)
(h)
silhouette 0.224
Δα (centred)
A_1
A_2
A_3
GIN hidden-state archetype ($k = 3$)
(i)
Spearman ρ = +0.68
Δα (centred)
S_g (rank, count-supported genes)
(j)
H+Mouse
5 species
H+M. fascicularis
H+M. mulatta
H+Pig
primates
density
Δα (centred)
(k)